

Assignment - 01.

Q1.

→ Python was developed by "Guido van Rossum" in 1981. He created it to make programming simple, readable.

Differences b/w C & Python:

i] C is compiler based, Python is Interpreter based

Q2.

→ Python's build process:

i] PVM's Interpreter reads the source code.

ii] It then converts the .py code into bytecode (.pyc)

iii] PVM executes the bytecode line by line.

Q3.

→ Bytecode is an intermediate form of python code generated after compilation.

It is usually stored in the --pycache-- folder as .pyc files.

Python uses bytecode because it speeds up program execution by avoiding recompilation every time.

Q4.

→ Python is both compiled & interpreted.

i] .py → compiled into bytecode

ii] PVM interprets → bytecode

Q5.

→ The --pycache-- folder stores compiled bytecode (.pyc) files.

Q6. Why is Python dynamically typed?

→ For ease, we don't need to declare a variable's data type. PVM determines it during execution.

Q7.

→ A] Platform Independent:

Same .py file can run on any windows / any OS without changing the code (as long as python is installed).

B] Portable.

Python programs can be moved from one system to another and still work with little or no modification.

Q8.

→ Python uses indentation to define code blocks instead of { }.

eg- if True:

print("Hi")

So after colon (:) next line starts with an indent.

Q9.

→ Python throws an error, `IndentationError`: It cannot identify the code block.

Q10.

→ 3 Programming Paradigms supported by Python:

- i] OOP
- ii] Functional (lambda)
- iii] Procedural.